
APMTH 50 Modeling Activity

1. Consider a uniform material with constant attenuation coefficient μ . Suppose light with intensity I_0 enters the material at $x = 0$. We'd like to get some intuition for the size of the output intensity I at $x = \Delta x$ relative to the input intensity I_0 .
 - (a) In general, do you expect I to be greater than, less than, or equal to I_0 ?
 - (b) Is it possible for $I = I_0$? When?
 - (c) Is it possible for $I = 0$? When?
2. Suppose it is observed that the rate of change of light intensity with respect to material thickness is proportional to the light intensity with the attenuation coefficient of the material as the proportionality constant. Write down a differential equation that reflects this observation. Solve the differential equation. Is the solution to the initial value problem (differential equation together with the initial condition at $x = 0$) consistent with your intuition?
3. Finally, suppose the material is nonuniform so that μ is no longer constant. In particular, suppose $\mu = \mu(x)$. Brainstorm with your group on how you might try to deal with this more realistic situation.